Blocco 3

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In the following hf will stand for hyperfinite and hhf will stand for hyperhyperfinite.

Esercizio 1. Define an equivalence relation on the Baire space by setting xEy iff for all large n, $x_n = y_n$. Prove that E is hf.

Proof. Consider $f : \omega^\omega \to 2^\omega$ that maps $a \in \omega^\omega$ in the sequence obtained by the following steps:

1. Write any element of a in binary
2. Put each element in a column of an $\omega \times \omega$ matrix
3. Scan the matrix diagonally (i.e. if M is the matrix, consider $\langle M(0,0), M(0,1), M(1,0), M(0,2), M(1,$

We claim f is a Borel reduction from E to E_0. This will be enough because E_0 is hf.

- $xEy \Rightarrow f(x)E_0f(y)$: Let M, N the matrix associated to x, y respectively. Since xEy we have that $\exists k \in \omega \forall n \geq k \forall m \ (M(n,m) = N(n,m))$. Let $s \geq k$ such that
$$\forall t \geq s \forall r \leq k M(r,t) = N(r,t) = 0$$
Then $\forall n \geq \frac{2s \cdot (2s+1)}{2} \left(f(x)(n) = f(y)(n) \right)$.
- $f(x)E_0f(y) \Rightarrow xEy$: Let $k \in \omega$ such that $\forall n \geq k \ (f(x)(n) = f(y)(n))$. Then
$$\forall n \geq k \ (x(n) = y(n))$$

$\square$

Esercizio 2. Show that $E \leq_B E'$ and E' is hhf, then E is also hhf.

Proof. Since E' is hf, we can write $E' = \bigcup_{n<\omega} E'_n$ with $\langle E_n : n < \omega \rangle$ increasing sequence of hf cbers. Let $f : X \to X$ be a Borel reduction from E to E'. For every $n < \omega$ define $xE_ny \iff f(x)E'_nf(y)$.

- $E \subseteq \bigcup_{n<\omega} E_n$:
$$xEy \Rightarrow f(x)E'f(y) \Rightarrow f(x)E'_nf(y) \text{ for some } n < \omega \Rightarrow xE_ny \Rightarrow x \bigcup_{n<\omega} E_n y \tag{1}$$
- $\bigcup_{n<\omega} E_n \subseteq E$:
$$x \bigcup_{n<\omega} E_n y \Rightarrow xE_ny \text{ for some } n < \omega \Rightarrow f(x)E'_nf(y) \text{ for some } n < \omega \Rightarrow f(x)E'f(y) \Rightarrow xEy \tag{2}$$

- for all $n < \omega$ E_n is hf cber :

  Fix $n < \omega$. Then E_n is a cber because f,E'_n are Borel and $E_n \subseteq E$. E_n is hf because f is a Borel reduction from E_n to E'_n and E'_n is hf.

- $\langle E_n : n < \omega \rangle$ is increasing :

  In fact $\langle E'_n : n < \omega \rangle$ is increasing, so we get

$$xE_ny \Rightarrow f(x)E'_nf(y) \Rightarrow f(x)E'_{n+1}f(y) \Rightarrow f(x)E'f(y) \Rightarrow xEy \tag{3}$$

We conclude observing that $E = \bigcup_{n<\omega} E_n$ witnesses E hf.

□

Esercizio 3. Show that the Cohen forcing $Add(\omega, 1)$ is weakly homogeneous.

Proof. Let p, q in $Add(\omega, 1)$ and let $p' \leq p, q' \leq q$ such that $dom(p) = dom(q), p'(n) = 0$ if $n \notin dom(p), q'(n) = 0$ if $n \notin dom(q)$. Now define $\pi : Add(\omega, 1) \to Add(\omega, 1)$ by setting for all $r \in Add(\omega, 1)$ and for all $n \in dom(r)$

$$\pi(r)(n) = \begin{cases} p'(n) \text{ if } r(n) = q'(n) \\ q'(n) \text{ if } r(n) = p'(n) \\ r(n) \text{ else} \end{cases}$$

Observe that π is an automorphism that can be extended to $\overline{\pi} : \overline{Add(\omega, 1)} \to \overline{Add(\omega, 1)}$. We conclude since $\overline{\pi}(p') = q' \not\perp q'$.

□

For the next problem, let k be a measurable cardinal with a normal measure U and let P be the Prickry forcing at k.

Esercizio 4. Prove that P is weakly homogeneous.

Proof. WLOG we can assume we are considering the version of P where for all $(s, A) \in P$ we have $max(s) < min(A)$. Let $p, q \in P$. Let $p' \leq p, q' \leq q$ such that $|stem(p')| = |stem(q')| = l$ and $p' = (s, A), q' = (t, A)$. Let G generic such that $p' \in G$. We want to find H generic such that $q' \in H$ and $M[G] = M[H]$. If $\vec{\alpha}, \vec{\beta}$ are the generic sequences associated to G, H respectively, then $M[G] = M[H]$ is equivalent to $|\vec{\alpha} \triangle \vec{\beta}| < \omega$. So it is enough showing that there exists $\vec{\beta}$ generic sequence such that coincide with $\vec{\alpha}$ on a tail and such that $stem(q') \subseteq \vec{\beta}$.

Consider $\vec{\beta}$ defined as

$$\vec{\beta}(n) = \begin{cases} stem(q')(n) \text{ if } n \leq l \\ \vec{\alpha}(n) \text{ else} \end{cases}$$

- $\vec{\alpha}, \vec{\beta}$ coincide on a tail : This is directly from definition.

- $\vec{\beta}$ extends $stem(q')$: This is directly from definition.

- $\vec{\beta}$ is a generic sequence : Consider

$$H = \{(x, B) \in P : x \text{ is initial segment of } \vec{\beta}, \big(\vec{\beta}(|x| + 1) \in B\big)\}$$

  It is clear that $\vec{\beta}$ is the generic sequence associated, but we need to verify H is generic. Let D dense and let $D' = \{(s \frown z, C) : (t \frown z, C) \in D\}$.

- $D' \neq \emptyset$ because D is dense.
- D' is dense under p': In fact let $(s \frown z, C) \leq p'$. By density of D there exists $(t \frown z \frown u, B) \in D$ with $B \subseteq C$. Then $(s \frown z \frown u, B) \in D'$.
- $G \cap D' \neq \emptyset$: In fact $p' \in G$, so one only needs to consider $\{r \in P : r \perp p'\} \cup D'$.

Let $(s \frown z, C) \in G \cap D'$. Then:

- $(t \frown z, C)$ is a valid condition because $max(s) = max(t)$
- $(t \frown z, C) \in D$ because $(s \frown z, C) \in D'$
- $(t \frown z, C) \in H$ because $\vec{\beta}$ extends $t \frown z$ and $\vec{\beta}(|t \frown z| + 1) \in C$ since $\vec{\beta}(|t \frown z| + 1) = \vec{\alpha}(|s \frown z| + 1) \in C$ (being G generic associated $\vec{\alpha}$).

So we have found $(t \frown z, C) \in H \cap D$

$\square$